

An Overview of Data Preprocessing for Short-Term Wind Power Forecasting

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Abstract

Wind power generation takes on an increasingly vital role in the power grid due to its environmental and economic benefits. However, the primary challenges that are related to the integration of wind power into power systems include variability, uncertainty. An accurate forecasting reduces operating costs and enhances power system stability. Wind power forecasting include many steps, including data collection, data preprocessing, the construction and training for models, and error calculation. Among them, data preprocessing plays an important role on the process of wind power forecasting since the inputs of the forecasting model would be sensitive to the quality of data. As a result, this paper presents a survey on the methods for wind-data processing. These methods aim to preprocess and extract suitable features from numerical weather prediction (NWP) wind speeds and measured wind power data. Finally, this paper used a case study to demonstrate the important of the preprocessing step on wind power forecasting.

Key words: Wind Power Generation, Forecasting, Data Preprocessing, Numerical Weather Prediction (NWP)

Introduction

Renewable energy such as wind or solar energy has attracted much attention among different renewable energy resources. However, the characteristic of wind speeds is intermittent, which brings a huge challenge on the operation of wind turbines. Therefore, forecasting wind power generation plays a vital role on power-related issues [1]. Some researches on wind power forecasts have offered credible estimates for comparing different prediction methods [2]. The uncertainty of wind generation characteristics affects the accuracy of wind power forecasting results [3]. Ensemble techniques considered weather uncertainty generated by different parameters of several weather models. The most significant inputs are the ensemble NWP wind speeds [4]. It is necessary to obtain the forecasting wind power generation in different time frames to reduce operational costs and improve power stability [5]. Data preprocessing includes cleaning, normalization, transformation, feature extraction, and feature selection. Some of these technologies can be applied to the research area about wind power forecasts. For instance, the process of data preprocessing has been applied to deal with the reduction or missing values of wind speeds or wind generation data.

This paper provides a complete literature review about the preprocessing techniques and related works for short-term wind power forecasts. This paper also provides a case study using a preprocessing technique – Kernel Principal Component analysis (KPCA). KPCA is utilized to extract the hidden features and identify useful information from the raw wind

inputs. Then, using KPCA obtains kernel principal components to reconstruct the phase space of time series of the inputs. After the data preprocessing of input variables, this work used XGBoost model to predict short-term wind power generation.

Literature Review and Background

A. Data Preprocessing (Data mining)

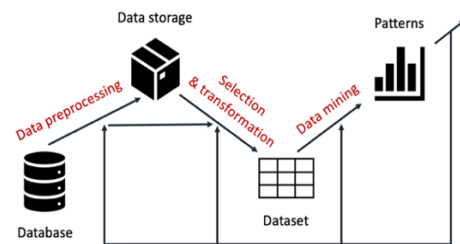


Fig. 1 Data preprocessing from database

Data mining is the process of discovering and revealing previously important, hidden, unknown, and valuable patterns in a database [6]. The process is shown in Fig. 1. Normally, data measured from wind farms contain outliers, noises, inconsistency or missing values. Therefore, the preprocess of input raw data improves the reliability and quality of forecasting performance. Data preprocessing techniques are divided into the following main parts:

- Data cleaning: it handles missing values or noises and removes unnecessary information from a database.
- Data integration: several data types collected from several databases are combined into a single type, causing redundant data, duplicate data or inconsistent data. Thus, an appropriate technique about data integration is significant.
- Data selection: in this step, relevant data are obtained from databases. Sometimes, data integration is performed before the process of data selection.
- Data transformation: it includes smoothing, aggregation, discretization, attribute construction, generalization, and normalization.
- Data reduction: if the data are large, data reduction is performed to obtain the most suitable subset of attributes. Data reduction techniques include data cube aggregation, dimensionality reduction, data compression and numerosity reduction. These techniques aim to map the data from their original dimension to lower dimension space while minimizing information loss.

B. Data preprocessing techniques

Several methods and algorithms have been developed in the field of data preprocessing. Data preprocessing can be divided into the following categories: classification, clustering,

regression analysis, and association rules [7]. In machine learning, classification is one of supervised learning techniques. It categorizes a given set of data into different classes; classification can be performed on both structured and unstructured data. Various classification algorithms have been developed, such as Logistic Regression, Naïve Bayes, Stochastic Gradient Descent, K-Nearest Neighbours, Decision Tree, Random Forest, Decision Tree, Random Forest, and Support Vector Machine [8]. Clustering is used to divide data into different groups based on similarity, density, intervals or particular statistical distributions [9]. Some examples for clustering are hierarchical, partitional, grid, density based, and model-based methods [10]. Regression analysis is the data mining procedure that identifies and analyzes the relationship among variables. It is widely used to define the probability distribution of a given variable [11]. Association rules are if-then statements that demonstrate the probability of interaction among data in large datasets [12]. Many algorithms for generating the association rules have been proposed; they include Apriori, Eclat or FP-Growth.

C. Numerical Weather prediction (NWP)

NWP models can represent atmospheric motion without requiring a substantial quantity of historical data, making them appropriate for forecasting wind power generation [13]. Despite significant improvements in the last few decades, NWP models frequently provide inaccurate wind-speed forecasts, particularly in a complex terrain. Wind power forecasting relies heavily on NWPs. Thus, the inputs of a wind power forecasting model should include NWP wind speeds. A typical process of wind power forecasting using ensemble NWP models may include a unique conversion model, multi-conversion forecasting model, or dimensional reduction model. For instance, forecasting

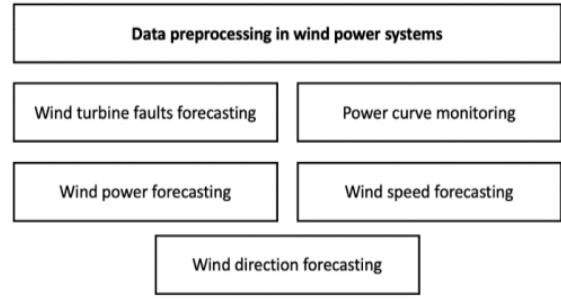


Fig. 4 Data preprocessing techniques in wind power systems

results based on both deterministic NWPs and ensemble NWPs can be achieved using either prediction error methodologies or direct methodologies [14].

For the past 30 years, Taiwan CWB has been developing global and regional NWP systems. Taiwan CWB has developed different Weather Research and Forecasting (WRF) modes, which are next generation mesoscale numerical weather prediction systems. These modes include deterministic WRF (WRFD), radar WRF (RWRF), and WRF ensemble prediction system (WEPS). In [14], the main features of NWP wind speeds are extracted from ensemble NWP data using a dimensional reduction approach that is based on Kernel density estimation; then a short-term probabilistic forecasting engine using LUBE-based neural network was constructed.

Related Works - Data preprocessing techniques in short-term wind power forecasting

Short-term wind power forecasts (from 30 minutes to 6 hours ahead) are used for economic dispatch and load

TABLE I
METHODOLOGIES IN SHORT-TERM WIND POWER FORECASTING

Ref	Proposed model	Input data	Lead time	Performance
[15]	K-means clustering based bad data detection module and neural network (NN)-based forecasting module	NWP Historical wind power	1 hour	RMSE all = 0.14
[17]	Principles of wavelet transform (WT), support vector machine (SVM) as well as characteristics of wind turbine generation system.	Wind speed, Wind power, Wind direction	1 hour	MRE = 10.23%, 7.05%
[18]	Data mining approach combining K-means clustering with Bagging Neural Network	Wind turbine parameters	1 hour	RMSE = 342.548 kW
[19]	Data preprocessing, adaptive neuro fuzzy inference system (ANFIS) model, post processing of predicted interval	NWP (spot, ensemble) wind speed Historical wind power	1 hour 3 hours	PICP, PINAW, CWC improved
[20]	Improve supervised forecasting model: SVM, ANN, Gaussian Processes (GPs). Data-driven feature extraction	Historical wind power Meteorological forecasting	1 hour	RMSE = 0.1377
[21]	Combining stacked autoencoders (SAE) with 3 hidden layers and back propagation (BP) algorithm	Historical wind power	15 mins	MAPE = 15.96%
[22]	Combination of neuro-fuzzy and ANN	Historical wind power	1 hour	MAPE = 10.46%
[23]	Genetic algorithm piecewise SVM model	Historical wind power Wind speed	1 hour	MRE = 10.88%
[24]	SVM, M5P tree, REP tree, Bagging Tree, Multilayer Perceptron	Historical wind power Wind speed	10 mins 1 hour	MAE = 5937 kW

Abbreviations: RMSE: Root Mean Square Error, MAPE: Mean Absolute Percentage Error, MRE: Mean Relative Error, PICP: Prediction Interval Coverage Probability, PINAW: Prediction Interval Normalized Average Width; CWC: Coverage Width Criterion.

management. Various data preprocessing applications in wind power systems are summarized in Fig. 4. However, because of its intermittency and unpredictability, wind power forecasting remains one of the most difficult tasks in wind power systems [16]. In this section, the studies about the application of data preprocessing techniques on wind power forecasting are summarized.

A. Data preprocessing for raw wind speeds

Reference [15] proposed a data preprocessing technique for raw wind speeds by using the empirical method. It includes the following steps: constant data processing, missing data processing, physical range check, data scaling, irrational and unnatural data processing. Because it is difficult to identify the unnatural and irrational data, the work in [15] utilized an unsupervised algorithm which called Local Outlier Factor (LOF) method to obtain and eliminate these unnatural and irrational data. The unsupervised learning algorithm can detect and categorize raw wind speeds based on the data's properties. In practice, it is simple and convenient to be implemented, especially when the operational data are unavailable.

Notably, in addition to wind related issues, the data preprocessing approaches can be used to a wide range of applications. For instance, the concept of weighted distance in preprocessing approaches may be also used to other outliers or cluster detection methods in other research fields.

B. Wind speed and wind power forecasts

Table 1 summarizes numerous approaches that were used in short-term wind power forecasting. It is observed that support vector machine and K-means clustering algorithm have offered a wide range of wind power forecasting applications. In a general review, ANN models based on wavelet transform or ANFIS models, as well as Stacked Autoencoders (SAE) models come into prominence. Furthermore, based on the literature review, if a data preprocessing technique is considered, the forecasting performance on wind power generation can be significantly improved. Through the preprocessing step, the important characteristics of wind speeds or wind power can be extracted, and certain unnecessary noises can be removed. In terms of input data, several works [17][19] used NWP data as input, while other works used historical wind data measured at wind farms. Additionally, other factors such as wind direction, air pressure, air temperature, humidity can be also considered because of their effects on wind power generation.

In short-term wind power forecasts, each approach has its unique characteristics and may provide effective results in different situations. ANFIS, ANN and Machine Learning techniques have been considered as the efficient and convenient models. Additionally, the techniques concerning about classification or clustering have been also applied to

wind power forecasting.

Many papers indicated that there is no single standard metric to evaluate the performance of the used forecasting models. Most researches use various metrics to calculate wind power forecasting errors. Among them, RMSE, MAPE and MAE are commonly used and can be utilized for making an appropriate evaluation for wind power forecasting.

Case study description

In this paper, real data were obtained from a 5.4 MW Hu-Si Wind Farm on Penghu Island, Taiwan. The Hu-Si Wind Farm has six wind turbines, and the capacity of each turbine is 0.9 MW. The inputs of the forecasting model include historical wind power data and NWP wind speeds. The above-mentioned WRFD and RWRF systems were used in this case study to improve the performance of wind power forecasts.

A. Methodologies

KPCA is an unsupervised technique that performs PCA in a kernel's feature space. KPCA might be used to reconstruct time series, and certain kernel principal components are selected to build the final phase space of nonlinear time series. The selection of kernel principal components is based on their correlations to the model output.

XGBoost is a popular decision tree-based ensemble learning approach that uses gradient descent architecture. First, a decision tree is selected as base learner. Each tree maps an input data point to a score in one of its leaves. In training process, XGBoost minimizes objective function that combines loss function and a penalty term for model complexity by building several trees. The trees are built sequentially such that each sub-tree aims to reduce the errors of the previous tree and then make the final prediction [25]. Furthermore, to reduce overfitting caused by having a large number of trees, stopping criteria for boosting is selected. To archive forecasting error, Root mean squared error (RMSE) is chosen as error measurement and calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P - P_0)^2} \quad (11)$$

where n is the number of data points; P_0 is the real historical power value; P is the wind power forecast value.

B. Forecasting Results

To demonstrate the superiority and the importance of data preprocessing. In this work, the results of wind power forecasting using KPCA-XGBoost model are presented. Then, the results with/without data preprocessing techniques are compared. In the developed forecasting model, the input data include historical wind power measurements and spot NWPs, and the spot NWPs include WRFD and RWRF. The collected dataset is separated into the training set (from July 2019 to January 2020) and the testing set (February 2020).

Fig. 6 shows the one-hour-ahead deterministic forecasting results using XGBoost, and the forecasting model utilized raw historical wind power values as input. Fig. 7 shows the forecasting results using XGBoost with KPCA preprocessing. In both figures, the blue line denotes forecasting values and the

TABLE III
ERROR IMPROVEMENT (RMSE)

Dataset	Without preprocessing	With preprocessing
Training	0.1780 MW	0.1258 MW
Testing	0.6102 MW	0.3708 MW

orange line denotes real values. It can be observed that the forecasting model with data preprocessing can track the actual data well. The forecasting errors from the two cases is summarized in Table III.

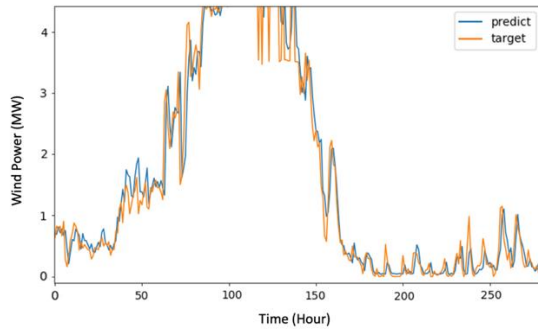


Fig. 6 Forecasting and real hourly wind power without KPCA

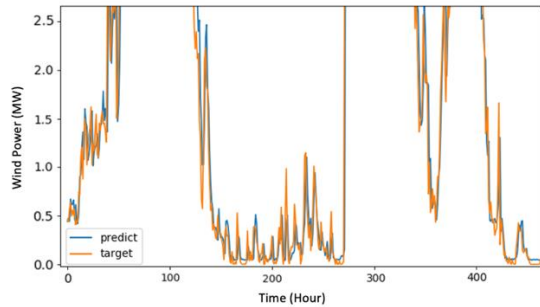


Fig. 7 Forecasting and real hourly wind power with KPCA

Conclusions

In this paper, various data preprocessing techniques were reviewed. In addition, the application of data preprocessing techniques on wind power systems or short-term wind power forecasts were also summarized in detail. Therefore, the purpose of this paper is to demonstrate the importance of data preprocessing on wind power forecasts. That is, using data preprocessing provides greater performance on wind power forecasting. Furthermore, A case study' results show that the proposed model improves forecasting accuracy, indicating the capability of KPCA on feature extraction. Therefore, the potential of machine learning techniques for wind power forecasting is high.

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